AQ9.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.
Limits:

Level 1:

Find the value of the following limit:

$\lim_{(x,\,y)\,\to\,(2,3)} \frac{x^2+y^2-y}{y^2-x^2}$lim(x,y)→(2,3) y2−x2x2+y2−y

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
1 point
Suppose ff and gg are multivariable functions defined on domain R^2R2 with the following limits:

$\lim_{(x,\,y)\,\to\,(0,0)} f(x,y) = l$ and$\lim_{(x,\,y)\,\to\,(0,0)} g(x,y) = k$lim(x,y)→(0,0)
f(x, y) = l and lim(x,y)→(0,0) g(x, y) = k

where $l, k \in R$l, k ∈ R. Which of the following options is(are) correct?

- [ ] $lim(x,y)\to(0,0)\ (f(x,y)+g(x,y)) = l+k$(x,y)→(0,0)lim (f(x, y) + g(x, y)) = l + k
- [ ] $lim(x,y)\to(0,0)\ (f(x,y).g(x,y)) = lk$(x,y)→(0,0)lim (f(x, y).g(x, y)) = lk
- [ ] $lim(x,y)\to(0,0)\ (5f(x,y)) \neq 5l$(x,y)→(0,0)lim (5f(x, y)) = 5l
- [ ] $lim(x,y)\to(0,0)\ (\frac{f(x,y)}{g(x,y)}) = \frac{l}{k}$(x,y)→(0,0)lim (g(x,y)f(x,y)) = kl, when $k \neq 0$k = 0

No, the answer is incorrect.
Score: 0
Accepted Answers:
$lim(x,y)\to(0,0)\ (f(x,y)+g(x,y)) = l+k$(x,y)→(0,0)lim (f(x, y) + g(x, y)) = l + k

$lim(x,y)\to(0,0)\ (f(x,y).g(x,y)) = lk$(x,y)→(0,0)lim (f(x, y).g(x, y)) = lk

$lim(x,y)\to(0,0)\ (\frac{f(x,y)}{g(x,y)}) = \frac{l}{k}$(x,y)→(0,0)lim (g(x,y)f(x,y)) = kl, when $k \neq 0$k = 0

Consider the function $f: R^2 \longrightarrow R$f : R2 ⟶ R defined as: $f(x,y) = \{\frac{4x^2y^2}{x^2+y^2}\ x, y \neq 0$ f(x, y) = {x2+y24x2y20x, y = 0x = y = 0What is the value of $lim(x,y)\to(0,0)\ (f(x,y))$(x,y)→(0,0)lim (f(x, y))?

No, the answer is incorrect.
Score: 0

Accepted Answers:
(Type: Numeric) 0
1 point
Suppose ff and gg are multivariable functions defined on domain R^2R2 as $f(x, y) = \frac{x^2 + y^2 + y(y - 2x - 1)}{x^2 - y^2}$f(x, y) = x2−y2x2+y2+y(y−2x−1) and $g(x, y) = \frac{2y^2 + x(x - 2y - 1)}{x^2 - y^2}$ g(x, y) = x2−y22y2+x(x−2y−1). Find the value of $lim(x,y) \to (1,1)\ (f(x, y) - g(x, y))$ (x,y)→(1,1)lim (f(x, y) − g(x, y)).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0.5
1 point
Find the value of the following limit:

$$\lim_{(x,\,y)\,\to\,(3,\,1)} \frac{x^2 - 3xy^2}{x^2 - 9y^4}$$(x,y)→(3,1)lim x2−9y4x2−3xy2

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0.5
1 point

Level 2:

∵ Consider the function $f: R^2 \longrightarrow R$f : R2 ⟶ R defined as:

$$f(x, y) = \begin{cases} \frac{3xy}{x^2 + y^2} & x, y \neq 0 \\ 0 & x = y = 0 \end{cases}$$f(x, y) = {x2+y23xy0x, y = 0x = y = 0

Use the given information to answer the questions 6-8:

If ff approaches to LL as (x, y)(x, y) approaches to the origin along Y-axis, then find the value of LL?
No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point
If ff approaches to LL as (x, y)(x, y) approaches to the origin along a straight line $y = 3x$y = 3x, then find the value of LL
No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0.9
1 point

1 point
Which of the following options is(are) false?
☐

$lim(x,y) \to (0,0)\ \ f(x, y) = 0$(x,y)→(0,0)lim f(x, y) = 0

☐
$lim(x,y) \to (0,0)\ \ f(x, y) = 1$(x,y)→(0,0)lim f(x, y) = 1

☐ Limit of the given function at (0,0) exists but is not equal to 0.
☐ Limit of the given function at (0,0) does not exist.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$lim(x,y) \to (0,0)\ \ f(x, y) = 0$(x,y)→(0,0)lim f(x, y) = 0

$lim(x,y) \to (0,0)\ \ f(x, y) = 1$(x,y)→(0,0)lim f(x, y) = 1

Limit of the given function at (0,0) exists but is not equal to 0.

1 point

Which of the following functions have limit value equal to 5 at the point (1,1,-1)?

☐
$f(x, y, z) = \frac{x + 3yz^2 - zx}{xy + yz + xz^2}$f(x, y, z) = xy+yz+xz2x+3yz2−zx

☐
$f(x, y, z) = \frac{3(x + y - z) + 7x^2 y + yz}{xyz + zy + xz}$f(x, y, z) = xyz+zy+xz3(x+y−z)+7x2y+yz

☐
$f(x, y, z) = x^3 + y^3 + z^3 - 2xy - 2yz - 4zx$f(x, y, z) = x3 + y3 + z3 − 2xy − 2yz − 4zx

☐
$f(x, y, z) = \frac{x^2 e^{\frac{x}{2}} + y^2 e^{\frac{y}{2}} + z^2 e^{\frac{-z}{2}}}{e^{\frac{x+y+z}{2}}}$f(x, y, z) = e2x+y+zx2e2x+y2e2y+z2e2−z

No, the answer is incorrect.
Score: 0
Accepted Answers:
$f(x, y, z) = \frac{x + 3yz^2 - zx}{xy + yz + xz^2}$f(x, y, z) = xy+yz+xz2x+3yz2−zx
$f(x, y, z) = x^3 + y^3 + z^3 - 2xy - 2yz - 4zx$f(x, y, z) = x3 + y3 + z3 − 2xy − 2yz − 4zx

1 point

Consider the function $f: R^2 \longrightarrow R$f : R2 ⟶ R defined as:

$f(x, y) = \{\frac{x^k y}{x^{2n} + y^{2n}} \quad x, y \neq 0 \quad 0 \quad x = y = 0$f(x, y) = {x2n+y2nxky0x, y = 0x = y = 0

where $k, n \in N \setminus \{0\}$k, n ∈ N \ {0}. Which of the following statements is(are) true about ff?

☐
If $k = 2n - 1$k = 2n − 1, then the limit at (0,0) exists and is equal to 0.

☐
If $k < 2n - 1$k < 2n − 1, then the limit at (0,0) does not exist.

☐
If $k > 2n$k > 2n, then the limit at (0,0) always exists and is equal to 0.

☐
If $k > 2n$k > 2n, then the limit at (0,0) does not exist.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If $k < 2n - 1$k < 2n − 1, then the limit at (0,0) does not exist.
If $k > 2n$k > 2n, then the limit at (0,0) always exists and is equal to 0.

Check Answers

Your score is: 0/10